Solar Si appendix

Sun · Si · downloadable appendix

Document	Solar Si appendix
Subject	Solar photospheric abundance appendix · Si
Author	Ryan Schmitt
Date	2026-10-18
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.278
Source	science commit 67a3cb4f0c66
Web	https://exoplanetcodex.org/systems/sol/elements/si/

Solar silicon: 7.512 ± 0.059

Asplund, Amarsi & Grevesse 2021: 7.51 ± 0.03 · this measurement +0.002 dex

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (Amarsi & Asplund 2017: nine Si I and one Si II line), 10 lines measured on one spectrum, each part weighted by its uncertainty.

Si I: 7.489 ± 0.074 · Kitt Peak (1984, molecfit) · VIS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Si II: 7.518 ± 0.155 · Kitt Peak (1984, molecfit) · VIS · Si II · Asplund et al. 2021 lines · 1D LTE

Si I: 7.577 ± 0.127 · Kitt Peak (1984, molecfit) · red-optical · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Highlighted Products

VIS

7.489 ± 0.074

±0.020 stat · ±0.071 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

7.577 ± 0.127

±0.036 stat · ±0.121 syst

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

7.522 ± 0.069

±0.026 stat · ±0.064 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.325 ± 0.104

±0.035 stat · ±0.098 syst

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · CRIRES+ H · Reference Grade

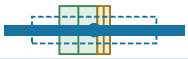
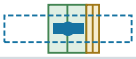
Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

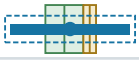
Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	38 products -- Si I · Asplund et al. 2021 lines; Si I · Deshmukh et al. 2022 lines; Si I · Elgueta et al. 2026 lines; Si II · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	24 products -- Si I · Codex-graded lines; Si II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS
A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ±0.142 stat ±0.098 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.511 ±0.024 stat ±0.100 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.514 ±0.047 stat ±0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.487 ±0.048 stat ±0.072 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.514 ±0.047 stat ±0.065 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.487 ±0.048 stat ±0.072 syst

VIS — IAG FTS
A(Si) dex; axis 7.15–7.83

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.519 ±0.094 stat ±0.103 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.489 ±0.105 stat ±0.101 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.511 ±0.015 stat ±0.070 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.484 ±0.016 stat ±0.076 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.511 ±0.015 stat ±0.070 syst

VIS — IAG FTS
A(Si) dex; axis 7.15–7.83

Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.484 ±0.016 stat ±0.076 syst

VIS — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.15–7.83

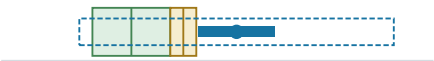
Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.528 ±0.144 stat ±0.102 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.491 ±0.311 stat ±0.101 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.513 ±0.020 stat ±0.065 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.486 ±0.021 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.513 ±0.020 stat ±0.065 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.486 ±0.021 stat ±0.071 syst

VIS — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.15–7.83

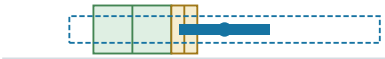
Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.518 ±0.110 stat ±0.110 syst
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.488 ±0.285 stat ±0.104 syst
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.516 ±0.019 stat ±0.064 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.489 ±0.020 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.516 ±0.019 stat ±0.064 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.489 ±0.020 stat ±0.071 syst

VIS axis: 7.15 · 7.32 · 7.49 · 7.66 · 7.83 A(Si) dex

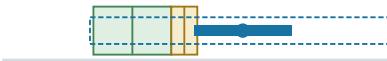
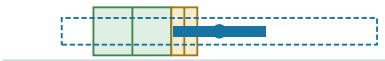
red-optical — IAG FTS
A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.591 ±0.030 stat ±0.121 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.573 ±0.028 stat ±0.125 syst

red-optical — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.41–7.74

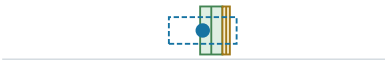
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.581 ±0.035 stat ±0.120 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.563 ±0.033 stat ±0.123 syst

red-optical — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.41–7.74

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.595 ±0.038 stat ±0.118 syst
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.577 ±0.036 stat ±0.121 syst

red-optical axis: 7.41 · 7.49 · 7.58 · 7.66 · 7.74 A(Si) dex

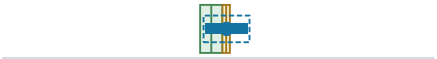
NIR — CRIRES+ J
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 2 lines		7.513 ±0.005 stat ±0.086 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.487 ±0.009 stat ±0.091 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 27 lines		7.527 ±0.061 stat ±0.083 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 27 lines		7.507 ±0.061 stat ±0.088 syst

NIR — CRIRES+ Y
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.018 stat ±0.064 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.018 stat ±0.071 syst
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 2 lines		7.457 ±0.011 stat ±0.082 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.430 ±0.002 stat ±0.087 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		7.582 ±0.073 stat ±0.086 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 4 lines		7.560 ±0.078 stat ±0.091 syst

NIR — CRIRES+ Y (full arm)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.018 stat ±0.064 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.018 stat ±0.071 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 8 lines		7.551 ±0.057 stat ±0.062 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 8 lines		7.523 ±0.059 stat ±0.069 syst

NIR — IAG FTS
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.554 ±0.121 stat ±0.070 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.542 ±0.121 stat ±0.076 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.528 ±0.088 stat ±0.135 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.497 ±0.125 stat ±0.077 syst

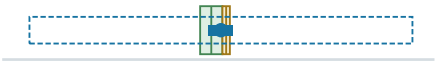
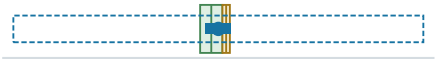
NIR — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.558 ±0.098 stat ±0.145 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.546 ±0.098 stat ±0.148 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.558 ±0.044 stat ±0.066 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.547 ±0.052 stat ±0.093 syst

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 3 lines		7.581 ±0.030 stat ±0.069 syst
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.552 ±0.035 stat ±0.075 syst
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 15 lines		7.536 ±0.027 stat ±0.057 syst

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 15 lines		7.522 ±0.026 stat ±0.064 syst
Si I · Codex-graded lines · 1D LTE Codex Grade · 46 lines		7.536 ±0.033 stat ±0.515 syst
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 43 lines		7.529 ±0.035 stat ±0.552 syst

NIR axis: 6.95 · 7.24 · 7.53 · 7.82 · 8.11 A(Si) dex

H — CRIRES+ H
A(Si) dex; axis 7.20–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.363 ±0.038 stat ±0.093 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.325 ±0.035 stat ±0.098 syst

H — CRIRES+ H (full arm, telluric-corrected)
A(Si) dex; axis 7.20–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.418 ±0.054 stat ±0.108 syst
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.380 ±0.047 stat ±0.112 syst

H axis: 7.20 · 7.30 · 7.39 · 7.49 · 7.59 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
VIS	Si I · Codex-graded lines; Si II · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
VIS	Si I · Asplund et al. 2021 lines; Si II · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
red-optical	Si I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	7.563–7.577 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.04
NIR	Si I · Codex-graded lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	4–46	7.497–7.560 (1D-NLTE)	telluric residual 0.55, line strengths (gf) 0.04
NIR	Si I · Deshmukh et al. 2022 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	7.428–7.552 (1D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
NIR	Si I · Elgueta et al. 2026 lines	CRIRES+ Y, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–15	7.430–7.546 (1D-NLTE)	line strengths (gf) 0.04, fit window 0.03
H	Si I · Elgueta et al. 2026 lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE	3	7.325–7.380 (1D-NLTE)	continuum placement 0.06, line strengths (gf) 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 59 lines.

Abundance outlier: disagrees with the other lines well beyond its error, and no solar analysis in the literature uses it.

Not measurable: the line falls in a data gap or a saturated telluric band.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Si II	3203.872	saturated (log W/λ = -4.18)
Si II	3210.028	saturated (log W/λ = -3.98)
Si II	3339.818	saturated (log W/λ = -4.04)
Si II	3856.017	saturated (log W/λ = -3.90)
Si II	3862.595	saturated (log W/λ = -4.15)
Si I	3905.523	saturated (log W/λ = -3.85)
Si I	4102.936	saturated (log W/λ = -4.10)
Si I	5948.541	saturated (log W/λ = -4.79)
Si I	7918.383	saturated (log W/λ = -4.88)

Species	Wavelength (Å, air)	Why it is excluded
Si I	7932.348	saturated (log W/λ = -4.81)
Si I	7944.001	saturated (log W/λ = -4.75)
Si I	8543.151	saturated (log W/λ = -4.23)
Si I	8648.465	saturated (log W/λ = -4.72)
Si I	8661.177	saturated (log W/λ = -4.27)
Si I	8728.010	saturated (log W/λ = -4.89)
Si I	8742.446	saturated (log W/λ = -4.90)
Si I	8752.007	saturated (log W/λ = -4.79)
Si I	8790.389	saturated (log W/λ = -4.82)
Si I	9021.590	saturated (log W/λ = -4.88)
Si I	9413.500	saturated (log W/λ = -4.56)
Si I	9887.047	abundance outlier
Si I	10313.197	abundance outlier
Si I	10371.263	saturated (log W/λ = -4.73)
Si I	10585.141	saturated (log W/λ = -4.46)
Si I	10603.425	saturated (log W/λ = -4.57)
Si I	10627.648	saturated (log W/λ = -4.85)
Si I	10660.973	saturated (log W/λ = -4.53)
Si I	10689.716	saturated (log W/λ = -4.70)
Si I	10694.251	saturated (log W/λ = -4.66)
Si I	10727.406	saturated (log W/λ = -4.62)
Si I	10749.378	saturated (log W/λ = -4.54)
Si I	10786.849	saturated (log W/λ = -4.59)
Si I	10827.088	saturated (log W/λ = -4.40)
Si I	10843.858	saturated (log W/λ = -4.73)
Si I	10868.790	saturated (log W/λ = -4.49)
Si I	10869.536	saturated (log W/λ = -4.34)
Si I	10885.333	saturated (log W/λ = -4.84)
Si I	10979.308	saturated (log W/λ = -4.62)
Si I	10982.058	saturated (log W/λ = -4.90)
Si I	11017.966	saturated (log W/λ = -4.68)
Si I	11187.588	saturated (log W/λ = -4.89)
Si I	11289.829	saturated (log W/λ = -4.86)
Si I	11611.060	saturated (log W/λ = -4.60)
Si I	11984.198	saturated (log W/λ = -4.46)
Si I	11991.568	saturated (log W/λ = -4.51)
Si I	12031.504	saturated (log W/λ = -4.38)
Si I	12103.534	saturated (log W/λ = -4.59)
Si I	12189.241	abundance outlier
Si I	12270.692	saturated (log W/λ = -4.62)
Si I	15077.332	saturated (log W/λ = -4.67)
Si I	15557.778	saturated (log W/λ = -4.88)
Si I	15888.409	saturated (log W/λ = -4.39)

Species	Wavelength (Å, air)	Why it is excluded
Si I	15960.063	saturated (log W/λ = -4.53)
Si I	16060.009	saturated (log W/λ = -4.64)
Si I	16163.691	saturated (log W/λ = -4.78)
Si I	16380.176	saturated (log W/λ = -4.72)
Si I	16381.534	saturated (log W/λ = -4.54)
Si I	16434.927	not measurable
Si I	17466.879	saturated (log W/λ = -4.45)

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Si appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 67a3cb4f0c66b8719fb12fe68fa9d992503d35eb
Feed Si.json v1.278 · updated 2026-10-17T23:20:32Z
Generated 2026-10-18T00:15:00-06:00 (source commit timestamp)
Feed SHA-256 7c1195f65f701ad6d0d6dbd97f55dee5cfdd7b07b6f7a1291b6ab2ce2619f708
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [6].

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [8].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [10].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [11].

Product-to-source index

VIS · HARPS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-18313ebf6e74

VIS · HARPS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-a600a1f3ca00

VIS · IAG FTS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7b7e31f61f2b

VIS · IAG FTS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-d538a2159904

VIS · Kitt Peak (Kurucz 2005) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-0d051b970724

VIS · Kitt Peak (Kurucz 2005) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-c60cd632806c

VIS · Kitt Peak (1984, molecfit) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-45b0eb7e0d7e

VIS · Kitt Peak (1984, molecfit) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-eb4fab3570b0

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-5005aa52ea70

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-c6dd0223e3f3

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-e636e6942aaf

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-50ccbe6c093c

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-78ada3f9735b

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-b8216b0c9359

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-d56ffa9cfc5d

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-6087fbe44c59

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-68a0da419260

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-fe06e8201b25

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-d01c0de482f8

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-bcdae8eb6758

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-02cc90577c3c

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-d9c04fe0bb0b

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-115af02fe503

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-e9b064f45645

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-f29c5165a798

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-ef230dcf5d1d

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-dab7a7917e06

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-319f483148d1

NIR · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-aa27575bc65f

NIR · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-703e8de44fc8

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-b1291f7f286b

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-f403a4693354

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-09438da36aef

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-63a93f8943d7

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-1a21e353cd5f

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-939560fefceb

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-01a5009be9be

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-724d64a49bde

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-a23c25d935d8

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-83f80d5ca2c1

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-b2f34d015eab

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-04b5965513ce

VIS · HARPS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-93a85c979084

VIS · HARPS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ea8e9f9add24

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-69325d3ca3fa

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ae649028ea9a

VIS · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-b8d5fe63d15d

VIS · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-4ac612585049

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-71ea831123e9

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-fab5c8319865

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-47f5784280f5

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-c16c0efa9866

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-e2f56f577f7c

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-a54926dacc0b

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-784a936bcab2

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-dcd2f1d1bede

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-e040cb5e6430

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-0b7df139857f

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-c31c1e8e3ffd

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-8fce3970b2cc

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-4eadb43adaf9

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ae5c932e8b0b

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-7ebb8d8816da

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-c8be7c1b1687

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-6171b736d381

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-b905b0754458

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

[3] — Amarsi, A. M.; Asplund, M. (2017). The solar silicon abundance based on 3D non-LTE calculations. MNRAS 464, 264-273 DOI: 10.1093/mnras/stw2445. source record

[8] — Elgueta, S.; Jofre, P.; Aguilera-Gomez, C.; Slumstrup, D.; Rojas Arriagada, A.; Heiter, U.; Casamiquela, L.; Zoccali, M.; Worley, C.; Soubiran, C. (2026). Gaia FGK benchmark stars: Selecting infrared lines for abundance determination. A&A 710, A111 DOI: 10.1051/0004-6361/202659148. source record

[11] — Deshmukh, S. A.; Ludwig, H.-G.; Kucinskas, A.; et al. (2022). The solar photospheric silicon abundance according to CO5BOLD: Investigating line broadening, magnetic fields, and model effects. A&A 668, A48 DOI: 10.1051/0004-6361/202142072. source record

Atomic data and gf authorities

[4] — Garz, T. (1973). Absolute oscillator strengths of Si I lines between 2500 A and 8000 A. A&A 26, 471-477 source record

[5] — O'Brian, T. R.; Wickliffe, M. E.; Lawler, J. E.; Whaling, W.; Brault, J. W. (1991). Lifetimes, transition probabilities, and level energies in Fe I. JOSA B 8, 1185 DOI: 10.1364/JOSAB.8.001185. source record

[7] — Kelleher, D. E.; Podobedova, L. I. (2008). Atomic Transition Probabilities of Silicon. A Critical Compilation. J. Phys. Chem. Ref. Data 37, 1285-1501 DOI: 10.1063/1.2734566. source record

[9] — Pehlivan Rhodin, A.; Hartman, H.; Nilsson, H.; Joensson, P. (2024). Accurate and experimentally validated transition data for Si I and Si II. A&A 682, A184 DOI: 10.1051/0004-6361/202245686. source record

Model and grid sources

[6] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[10] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record